# UNIVERSITY OF PISA

## Department of Information Engineering

**Master's Degree Program in Biomedical Engineering**

**"Multifunctional and Reconfigurable Metasurfaces for Wireless Power Transfer in Active Implantable Medical Devices"**


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## 1. Context, Problem, and Objectives

Active Implantable Medical Devices (AIMDs) predominantly rely on primary non-rechargeable batteries characterized by an operational lifetime of 6–10 years. Battery depletion necessitates invasive surgical intervention to replace the entire unit, exposing patients to surgical risks, systemic infection hazards, and significant economic burdens. The clinical market for these platforms is expanding rapidly due to demographic aging and the rising incidence of chronic cardiovascular pathologies. The core drivers transforming this sector are miniaturization, high-bandwidth connectivity, and wireless power transmission.

Wireless charging completely removes the volumetric penalty of bulky batteries and the need for periodic surgical revisions; however, emerging therapeutic applications require motile or sensorized implants that directly challenge the conventional constraints of Inductive Power Transfer (IPT), namely operational distance and spatial misalignment. Due to anatomical distances and the small form factor of the implanted receiver, the inductive coupling coefficient $k$ is inherently weak ($k < 0.1$). Consequently, the overall Power Transfer Efficiency (PTE)—which scales quadratically with $k$ ($PTE \propto k^2$)—remains severely limited.

To address these constraints, magnetic metasurfaces (MTS)—planar arrays of near-field resonant elements placed between the primary transmitter (Tx) and secondary receiver (Rx)—have emerged as a compelling solution to concentrate and redirect the magnetic flux toward the load. When actively controlled via voltage-actuated reactive components at the unit-cell level, these surfaces can synthesize localized high-field regions (dynamic hotspotting), dynamically tracking the spatial displacement of the receiver. Furthermore, magnetic metasurfaces have been proven to preserve their electromagnetic response under mechanical deformation, enabling flexible, conformal implementations compatible with wearable anatomical skin patches.

This landscape makes the development of a framework capable of energizing motile implants both robustly and non-invasively of paramount engineering value. To the best of our knowledge, this work demonstrates for the first time a unified system capable of both **sensorless spatial tracking** of an implanted receiver and **closed-loop dynamic power transfer**.

- **Specific Engineering Objectives:**
    1. Achieve sensorless, passive spatial localization of the receiver (Rx);
    2. Implement a planar, low-profile architecture integrable into conformal wearable patches;
    3. Experimentally validate functional power transfer under dynamic tracking while maintaining compliance with human safety standards.
- **Proposed Solution:** A dual-layer magnetic metasurface that focuses near-field magnetic energy using voltage-controlled varactors, governed by an acquisition routine that reconstructs Rx position and shifts the focal spot in real time.

## 2. Theoretical Framework and State of the Art

Inductive power transfer operates within the magneto-quasi-static regime, enabling lumped-parameter equivalent circuit modeling. Magnetic metasurfaces are realized as arrays of near-field resonant RLC meta-atoms. Despite the high dimensionality of multi-element arrays, analytical homogenization allows the composite surface to be modeled as an equivalent lumped resonator.

Coupled with the primary transmitter and secondary receiver, the complete assembly is characterized by a $3 \times 3$ impedance matrix with source-dependent complex magnetic permeability. The effective mutual impedance:

$$Z_{12}^{eff} = j\omega M_{12} + \frac{\omega^2 M_{13} M_{23}}{Z_{33}}$$

achieves maximum power transfer at self-resonance. This analytical model enables precise calculation of unit-cell reactive loading to enforce target current profiles, ensuring reliable power delivery to misaligned loads.

Biological tissue layers interact strongly with propagating and reactive electromagnetic fields. Tissue dielectric properties were modeled incorporating Schwan dispersion and Gabriel's 4-pole Cole-Cole formulation. The trade-off between receiver coil miniaturization and tissue dissipation losses (which escalate rapidly at higher frequencies) dictated the selection of the **13.56 MHz ISM band** for high-efficiency power delivery.

- **State-of-the-Art Benchmarking:**
    - Aboualalaa and Pokharel demonstrated a 3-layer CSRR metamaterial operating at 280 MHz (total PTE ~53% at 20 mm, peak SAR 1.6 W/kg), but with a fixed, static focal region;
    - Li et al. utilized genetic algorithms to optimize passive metasurfaces (PTE 2.6% at >50 mm, +56% relative to unassisted links), lacking spatial tracking;
    - Ye et al. implemented dynamic reconfiguration via gradient descent on received load power, requiring active telemetry and complex circuitry on the implant side;
    - Falchi et al. validated varactor-tuned metasurfaces using manual parameter sweeps.

- Existing literature lacks a design that achieves **sensorless localization without placing active computational or telemetry burdens on the implanted capsule**.

## 3. Architecture Selection: In Silico Modeling (FEKO MoM + MATLAB)

Numerical modeling was conducted using the Method of Moments (MoM) in Altair FEKO. For each candidate geometry, the uncompensated impedance matrix of the raw multi-coil array was computed. Subsequently, in MATLAB, reactive matching networks were synthesized to tune resonance through series capacitive loading. This decoupled modeling methodology enabled rapid parametric sweeps across multiple operational frequencies without recalculating full-wave electrodynamic meshes.

Computational benchmarking identified the **Double Layer Dual Band (DD)** topology as the optimal design. This architecture isolates sensing from power transfer at the physical hardware layer by employing two independent metasurfaces resonant at distinct frequencies:

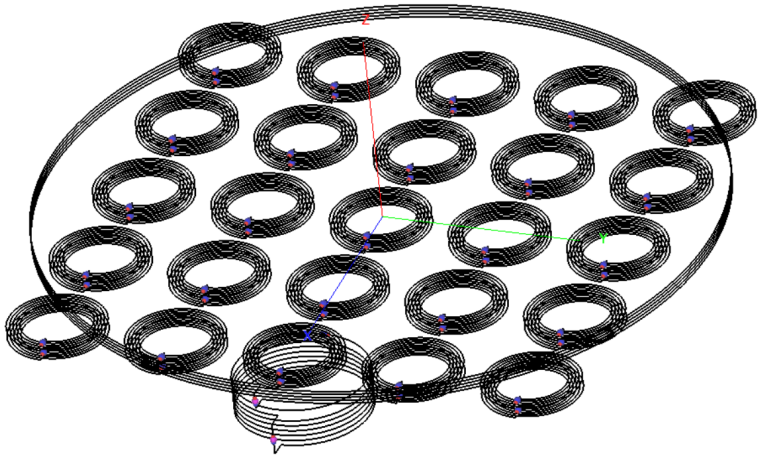


**Figure 1.** *Schematic concept of the Double Layer Dual Band (DD) metasurface architecture, showing the decoupled 160 MHz sensing layer (Layer 1) and the 13.56 MHz power delivery layer (Layer 2).*

- **Layer 1 (Sensing & Spatial Tracking — 160 MHz):** Operates on the propagation of magneto-inductive (MI) waves generated via nearest-neighbor mutual inductance governed by Lenz's Law. The passive implant behaves as a localized magnetic perturbation ("defect") that alters the spatial distribution of the MI wave. Two actively driven "pivot" resonators positioned orthogonally on the Cartesian perimeter launch the sensing wave.
- **Layer 2 (Power Transfer & Dynamic Focusing — 13.56 MHz):** Dedicated to high-efficiency power delivery. It focuses magnetic flux toward the receiver by dynamically biasing a reconfigurable $3 \times 3$ sub-matrix of unit cells aligned with the detected position. Physical layer decoupling prevents high-frequency sensing noise from coupling into the power transfer stage and eliminates resistive dissipation on the charging path.

- **Operating Principle:** The passive receiver establishes localized frequency splitting and distinct phase shifts between the pivot resonators. By matching the sampled transmission spectrum against a pre-calibrated Look-Up Table (LUT), the coordinates of the implant are mapped deterministically while keeping the implant completely passive.

## 4. Computational Dosimetric Safety (SAR)

A homogeneous muscle tissue phantom ($160mm \times 30mm$, $\epsilon_r = 143.5$, $\sigma = 0.623 S/m$ at 13.56 MHz) was implemented utilizing the 4-pole Cole-Cole dispersion model from the IFAC database. The phantom was interposed between the primary transmitter and the implant receiver, with verification methodologies structured under the **ASME V&V 40-2018** standard.

Under continuous single-frequency excitation at 13.56 MHz with 1 W input power, the computed 10-g averaged Specific Absorption Rate (SAR) was **0.76 W/kg**. This remains comfortably below the regulatory limit of **2.0 W/kg** established by **ICNIRP** guidelines and **IEEE C95.1** standards, proving inherent biological safety.

## 5. Hardware Prototyping & PCB Implementation

The complete physical prototype was fabricated on standard FR4 substrate (thickness: 0.8 mm, copper weight: 35 µm). All surface-mount passive components, varactors, and connectors were populated and soldered manually:

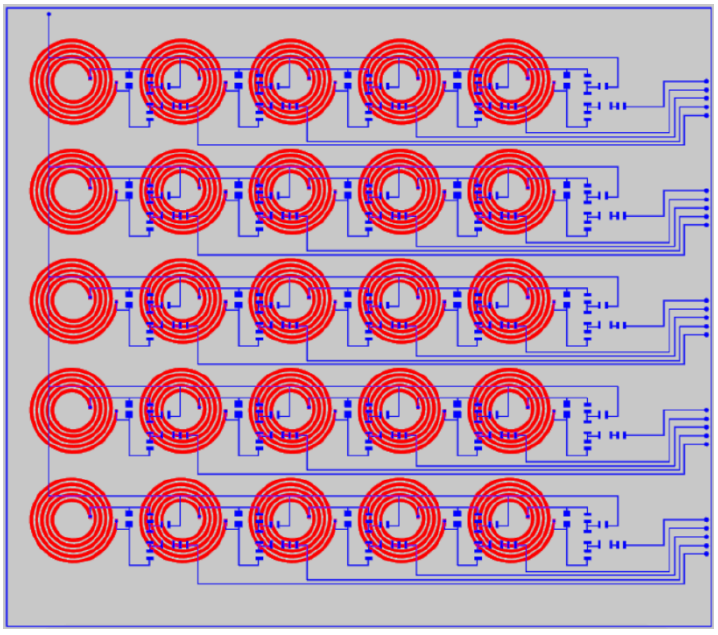


**Figure 2.** *Gerber fabrication layout of the 13.56 MHz reconfigurable power transfer layer (Layer 2), detailing the* $5 \times 5$ *planar resonator array, varactor bias lines, and tuning networks.*

- **Primary Transmitter (Tx):** Planar spiral coil with 2–10 pF tuning trimmer ( $C_{nom} = 8.27pF$);
- **WPT Receiver (Rx):** Resonant coil with 174 pF nominal tuning capacitance (150 pF fixed C0G dielectric + 8–45 pF variable trimmer; self-resonance at 240 MHz);

- **Sensing Receiver:** Decoupled loop tuned to 160 MHz ($C = 18.4pF$);
- **Sensing Metasurface (Layer 1):** $5 \times 5$ cell array tuned to 160 MHz, integrated with two perimeter pivot elements;
- **Power Metasurface (Layer 2):** $5 \times 5$ cell array equipped with reverse-biased varactor diodes, driven by an Arduino microcontroller through a 25-channel Digital-to-Analog Converter (DAC).

Power Transfer Efficiency was calculated directly from measured scattering parameters:

$$\eta = \frac{|S_{21}|^2}{1-|S_{11}|^2}$$

In this experimental validation setup, the WPT power loop and the sensing loop were implemented as decoupled physical receivers.

## 6. Experimental Characterization and System Validation

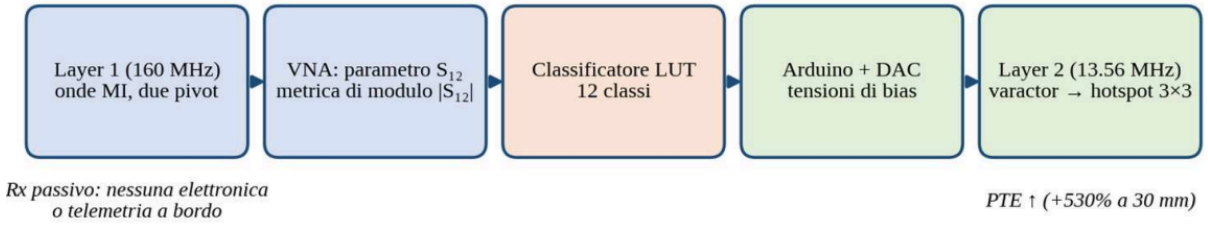



**Figure 3.** *System control loop execution sequence: from 160 MHz magneto-inductive sensing on Layer 1, through VNA spectral acquisition and LUT classification, to Arduino/DAC varactor biasing on Layer 2.*

- **Calibration & LUT Clustering:** All Layer 2 cells were calibrated to resonate at 12.8 MHz under a 0 V bias state ($C_{var} \approx 50pF$ combined with 330 pF fixed parallel capacitance). Spectral characterization was conducted using a calibrated Keysight FieldFox 2-port RF Vector Network Analyzer (100 kHz – 6 GHz). The baseline classification space required 25 repeated acquisitions across 25 spatial grid coordinates (625 acquisitions total). To maximize classification repeatability, a spatial clustering scheme was implemented, condensing the LUT into **12 discrete spatial classes** (5 primary $3 \times 3$ sub-matrices plus 7 boundary cells). The transmission magnitude $|S_{!2}|$ was adopted as the primary tracking metric.

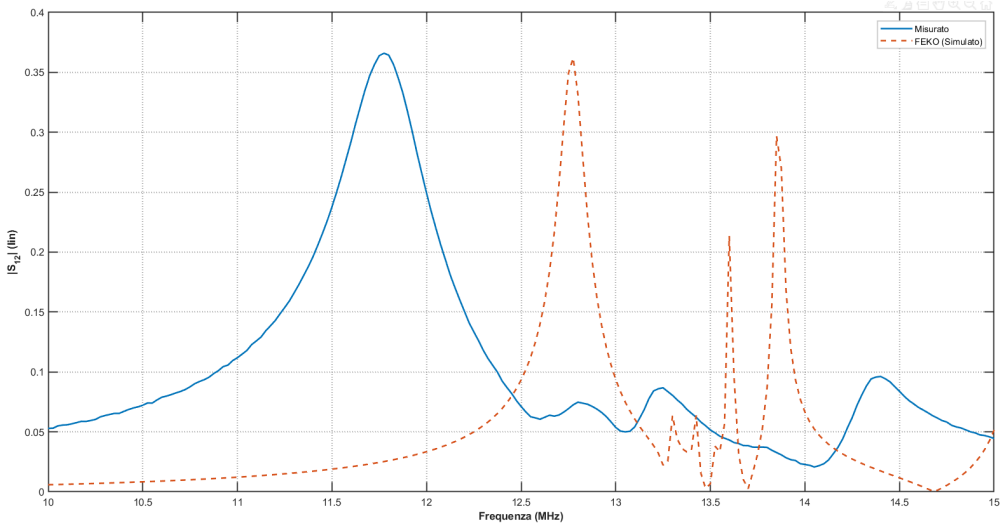



**Figure 4.** *Measured vs. simulated transmission coefficient spectrum ($|S_{12}|$) through the tissue phantom, experimentally confirming the three characteristic resonant peaks predicted in silico.*

- **Localization Accuracy (Test T1):** Evaluated via 10-repeated 5-fold cross-validation using Root Mean Square Error (RMSE) and Pearson correlation. Mean classification accuracy reached **~92%** (92.3% and 91.9% over 500-point spectra), proving robust against spectral decimation down to 10 points per sweep. Because classification errors remain confined within the spatial boundary of the same $3 \times 3$ sub-matrix, the effective operational tracking accuracy approaches **100%**.

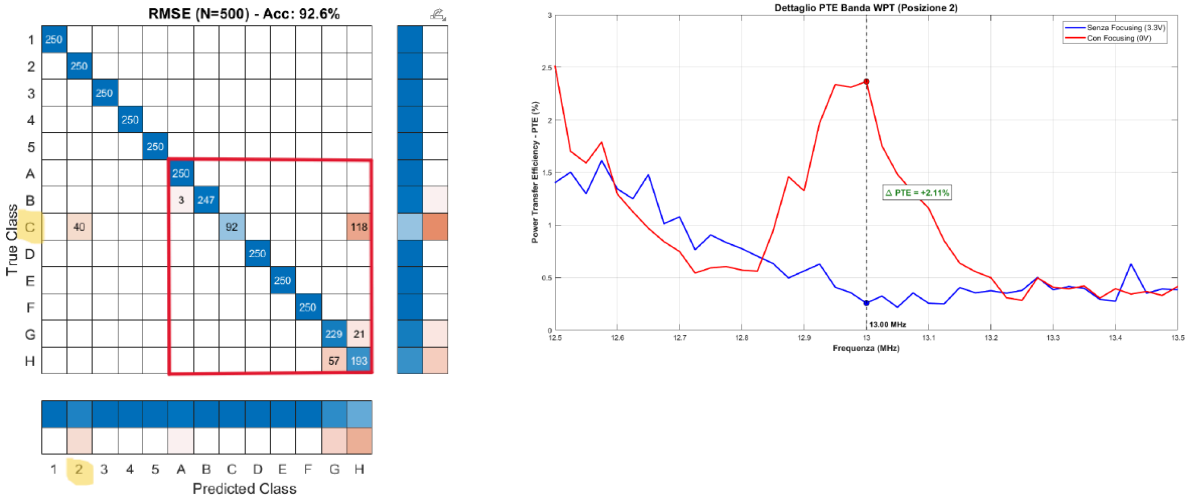



**Figure 5.** *Test T1 experimental findings: (a) 12-class spatial confusion matrix validating tracking precision; (b) Measured Power Transfer Efficiency vs. frequency demonstrating performance enhancement via dynamic focusing.*

- **Dynamic Focusing Gain (Test T1):** At a separation distance of 30 mm, measured PTE increased from **0.4% (unfocused baseline) to 2.4% (actively focused)**, corresponding to a **+530% relative improvement** (consistent with the +600% enhancement predicted in silico). Realized resonance shifted slightly to ~13 MHz due

to passive component tolerances. Applying a simplified 5-hotspot LUT instead of 9 preserved an average PTE of 1.39% (a minor reduction under 13% relative to the ideal configuration).

- **System Latency Profile:** Total closed-loop turnaround time was estimated conservatively at **~116 ms** (VNA acquisition & TCP overhead ~16 ms; algorithm execution ~0.2 ms; USB-UART transmission at 115200 baud ~100 ms). The latency budget is currently dominated by benchtop test instrumentation.
- **Biological Safety Validation (Test T2):** An agar-based muscle-mimicking phantom (2% w/v agar, 0.34% w/v NaCl) was fabricated per **IEC/IEEE 62209-1528** guidelines. Measured transmission spectra exhibited the identical three resonant peaks computed in simulation, providing rigorous indirect physical validation of the numerical model and confirming the **0.76 W/kg SAR safety margin**.

## 7. Benchmarking, Limitations, and Future Developments

### Synoptic Benchmarking Table

| Study / Reference | WPT / Sensing Frequency | Metasurface Architecture | Spatial Tracking / Distance | Implant Electronics Complexity | Power Transfer Efficiency (PTE) |
|---|---|---|---|---|---|
| **Aboualalaa et al. [20]** | 280 MHz / None | 3-Layer CSRR (MNZ) | Static Focus / 20 mm | Passive | 53.0% (1.6 W/kg peak SAR) |
| **Li et al. [11]** | 6.78 MHz / None | Stacked Multi-layer | Static / > 50 mm | Passive | 2.6% (+56% vs. passive baseline) |
| **Ye et al. [6]** | 13.56 MHz / RF Telemetry | 1-Layer Reconfigurable | Active Gradient Descent / N/A | Active RF Sensor & Telemetry | N/A (+180% $\| S_{21} \|$) |
| **Falchi et al. [22]** | 6.78 MHz / None | 1-Layer Reconfigurable | Manual Sweeps / 30 mm | Passive | Not Reported |
| **This Work (DD)** | **13.56 MHz / 160 MHz (MI)** | **Double Layer Dual Band** | **Sensorless LUT / 30 mm** | **Zero-Power Passive Receiver** | **2.4% (+530% via Dynamic Focusing)** |

- **Key Engineering Contributions:**
    - Achieves autonomous spatial tracking and localized beamforming without requiring sensors, batteries, or active RF communications on the implant side, avoiding the "dead battery trap".
    - Provides a 5-to-6-fold efficiency increase relative to unfocused metasurfaces while keeping SAR well below regulatory thresholds.
    - Introduces the **Double Layer Dual Band (DD)** topology, physically isolating the high-frequency sensing wave (160 MHz) from high-power delivery (13.56 MHz).
    - Demonstrates sensorless localization via magneto-inductive (MI) wave perturbation using two orthogonal pivot cells and an optimized 12-class LUT.
    - Employs an integrated design verification methodology combining ISO 14971/MDR risk management, computational SAR modeling, and biological phantom validation.
- **Declared Technical Limitations:**
    - The 5-to-6-fold gain is measured relative to the identical unfocused metasurface, not against UHF designs (e.g., [20]), which operate at fundamentally different frequencies, distances, and PTE metrics. Absolute PTE (2.4% at 30 mm without rectification) remains modest as RF-DC rectification was excluded from the project scope.
    - Susceptibility to ambient parasitics, RF cabling, and passive component tolerances (~13.56 MHz shifting to ~13 MHz) in unshielded bench environments. Spatial classification maps into 12 discrete zones rather than continuous millimeter coordinates.
    - Biological safety validation remains indirect via spectral matching; closed-loop latency was estimated analytically rather than captured end-to-end.
- **Future Work:**
    - Migration to an integrated embedded controller to reduce closed-loop latency below 5 ms;
    - Realization of conformal designs on flexible substrates (polyimide/PDMS) for anatomical skin patches;
    - Dynamic impedance matching incorporating complex current weighting coefficients;
    - Miniaturization of combined concentric WPT/sensing coils on the implant package.

## Conclusion

The Double Layer Dual Band (DD) metasurface architecture successfully demonstrates proof-of-concept sensorless tracking and dynamic magnetic near-field focusing for moving implanted loads while guaranteeing compliance with clinical SAR safety regulations. This research establishes a solid framework for wearable, conformal patches capable of transcutaneous wireless recharging for next-generation active implantable medical devices.